Hyun-Sik Jeon

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Research Interest | *VLM, Spatial Reasoning, 3D Visual Grounding, Autonomous Driving, Multi-Modal, Model Lightweighting*

Education

Seoul National University of Science and Technology (SeoulTech) Seoul, Republic of Korea
M.S. in Automotive Engineering Mar 2024 – Feb 2026

- Thesis Advisor: *Prof. Jong-Eun Ha*
- Thesis title: Spatial Understanding for Camera-based Autonomous Driving Perception and 3D Visual Grounding
- GPA: 3.81 / 4.5 (3.39 / 4.0)

Shinhan University Gyeonggi-do, Republic of Korea
B.S. in Future Automotive Engineering Mar 2018 – Feb 2024

- GPA: 3.63 / 4.5 (3.23 / 4.0), Major GPA: 3.83 / 4.5 (3.40 / 4.0)

Publications and Presentations

1. **H.-S. Jeon**, J.-E. Ha*, "Enhancing Spatial Reasoning of SpatialRGPT Through a Visual Saliency Branch," *Robotics and Autonomous Systems*, in submission, 2026.
2. S.-H. Kang, **H.-S. Jeon**, J.-E. Ha*, "Robust Zero-Shot 3D Visual Grounding via Structural Query Decomposition and Dynamic Viewpoint Selection," *Multimedia Tools and Applications*, under revision, 2026.
3. **H.-S. Jeon**, S.-H. Kang, J.-E. Ha*, "Enhancing SeeGround with Relational Depth Text for 3D Visual Grounding," *Applied Sciences*, vol. 16, no. 2, p. 652, 2026. (SCIE)
4. **H.-S. Jeon**, Y.-M. Kim, J.-E. Ha*, "Object Mask Module for Enhancing Multi-view 3D Occupancy Perception Performance Based on ViewFormer," *Journal of Institute of Control, Robotics and Systems (ICROS)*, vol. 31, no. 10, pp. 1160–1168, 2025. (in Korean) (SCOPUS)
5. **H.-S. Jeon**, S.-M. Park, J.-E. Ha*, "UniAD Model Lightweighting and Performance Comparison," *Journal of Institute of Control, Robotics and Systems (ICROS)*, vol. 31, no. 4, pp. 256–264, 2025. (in Korean) (SCOPUS)
6. J.-W. Lee, **H.-S. Jeon**, Y.-S. Choi*, "Steering and Velocity Control Between Traffic Cones for a LiDAR-based Autonomous Driving Platform in Low-spec Environments," *Korea Institute of Information & Telecommunication Facilities Engineering (ITFE) Conference*, pp. 227–228, 2023. (in Korean)
7. J.-W. Lee, **H.-S. Jeon**, Y.-S. Choi*, "Using a LiDAR Sensor for Autonomous Driving in a Low-end Environment," *The Institute of Electronics and Information Engineers (IEIE) Conference*, pp. 2434–2436, 2023. (in Korean)
8. S.-J. An, S.-T. Kim, **H.-S. Jeon**, Y.-S. Choi*, "A Method for Improving Lane Recognition by Adaptive Removing Vehicle Elements," *Korea Institute of Information & Telecommunication Facilities Engineering (ITFE) Conference*, pp. 24–25, 2022. (in Korean)
9. **H.-S. Jeon**, S.-J. An, S.-T. Kim, Y.-S. Choi*, "Vehicle Element Removal Method to Improve Lane Recognition," *The Institute of Electronics and Information Engineers (IEIE) Conference*, pp. 80–82, 2022. (in Korean)

Research Experiences

Artificial Intelligence System (AIS) Lab Mar 2024 – Feb 2026
SeoulTech University, *Advisor: Prof. Jong-Eun Ha*

- Relational Depth Text: +3.54%p on Nr3D with a 7B VLM; ViewFormer occupancy; UniAD lightweighting.

Undergraduate Researcher Jun 2022 – Dec 2023
Shinhan University, *Advisor: Prof. Yong-Soo Choi*

- Co-authored 4 papers (1 as first author) on vision/LiDAR-based perception for low-spec autonomous platforms.

Projects

Dynamic Scene Understanding via Integration of Neural Rendering and Transformers (Ongoing Lab Project) Mar 2024 – Feb 2026
SeoulTech University, *Advisor: Prof. Jong-Eun Ha*

- Mid-Career Research Program, National Research Foundation of Korea (NRF), Ministry of Science and ICT (MSIT)

Professional Experiences

Field Practicum Trainee, Ways1 Inc. Jun 2025 (5 days)
Supervisor: Jin-woo Kim

- Prototyped a GAT-based E2E state estimation model to replace the Kalman filter
- Keywords: Graph Attention Networks (GAT), E2E State Estimation, PyTorch Geometric

Founder & Vision Team Leader, Autonomous Driving Club "S.D.S" Mar 2022 – Feb 2024
Shinhan University, *Advisor: Prof. Yong-Soo Choi*

- Participant, 2023 University Creative Mobility Competition | MOLIT, KATRI, KASA – Oct 2023
- Keywords: Vision-based Perception System, End-to-End Training Pipeline, Custom Traffic Dataset Curation

Teaching Experiences

Teaching Assistant | *SeoulTech* Fall 2024 – Fall 2025
Automotive Structure, Basic Lab, and Automotive Combustion Engineering

- Instructor: Prof. Su-Ryong Lee

Honors and Awards

Research Publication Grant (SCOPUS Paper) | *SeoulTech* Jul 2025
National Advanced Robotics Training Fellowship | *MOTIE & KIAT* Jun 2024 – Feb 2026
Future Talent Scholarship for Research Excellence | *SeoulTech* Jul 2024

Military Service (Mandatory)

Republic of Korea Air Force Feb 2020 – Nov 2021
Sergeant, Vehicle Maintenance Specialist | *Completed*

Skills

ML/DL Frameworks: PyTorch, MMCV, PaddlePaddle
Robotics & Sim: ROS2, IsaacGym
Programming: Python, C++, C, MATLAB, LaTeX
Platforms & Environments: Linux (Ubuntu), Windows, Docker, Anaconda, Git/GitHub, Hugging Face
Mechanical Engineering Tools: CAD, CATIA, Abaqus, PAM-STAMP
Languages: Korean (Native), English (Proficient)